

Структура научной презентации

Простейший вариант

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Раздел 1

Цель работы

Здесь приводится формулировка цели лабораторной работы. Формулировки цели для каждой лабораторной работы приведены в методических указаниях.

Цель данного шаблона — максимально упростить подготовку отчётов по лабораторным работам. Модифицируя данный шаблон, студенты смогут без труда подготовить отчёт по лабораторным работам, а также познакомиться с основными возможностями разметки Markdown.

Раздел 2

Задание

Здесь приводится описание задания в соответствии с рекомендациями методического пособия и выданным вариантом.

Раздел 3

Теоретическое введение

Теоретическое введение

Здесь описываются теоретические аспекты, связанные с выполнением работы. Например, в табл. 1 приведено краткое описание стандартных каталогов Unix.

Таблица 1: Описание некоторых каталогов файловой системы GNU Linux

Имя каталога	Описание каталога
/	Корневая директория, содержащая всю файловую
/bin	Основные системные утилиты, необходимые как в однопользовательском режиме, так и при обычной работе всем пользователям
/etc	Общесистемные конфигурационные файлы и файлы конфигурации установленных программ
/home	Содержит домашние директории пользователей, которые, в свою очередь, содержат персональные настройки и данные пользователя
/media	Точки монтирования для сменных носителей
/root	Домашняя директория пользователя root

Раздел 4

Выполнение лабораторной работы

Раздел 5

Экспоненциальный рост

Цель: Исследовать решение уравнения $du/dt = \alpha u$.

Инициализация проекта и загрузка пакетов

Уравнение экспоненциального роста: $du/dt = \alpha u$, $u(0) = u_0$

Первый запуск с параметрами по умолчанию

Зададим начальные параметры:

Построим график решения:
Сохраним график в папку plots

Создадим таблицу с данными:
Вычислим удвоение популяции:

Сохранение всех результатов

This page was generated using Literate.jl.

```
Chocolatey v2.6.0
PS C:\WINDOWS\System32> choco install git
Chocolatey v2.6.0
Installing the following packages:
git
By installing, you accept licenses for the packages.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading chocolatey-compatibility.extension 1.0.0... 100%
```

Рисунок 1: Установка git

Интерфейс командной строки

```
PS C:\WINDOWS\System32> choco install gh -y
Chocolatey v2.6.0
Installing the following packages:
gh
By installing, you accept licenses for the packages.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading gh 2.86.0... 100%
|
```

Рисунок 2: Установим интерфейс командной строки к github:

```
Chocolatey installed 1/1 packages.  
See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).  
PS C:\WINDOWS\system32> git config --global user.name "Mehmet efe kantoz"  
PS C:\WINDOWS\system32> git config --global user.email "1032228428@pfur.ru"  
PS C:\WINDOWS\system32>
```

Рисунок 3: Зададим имя и email владельца репозитория:

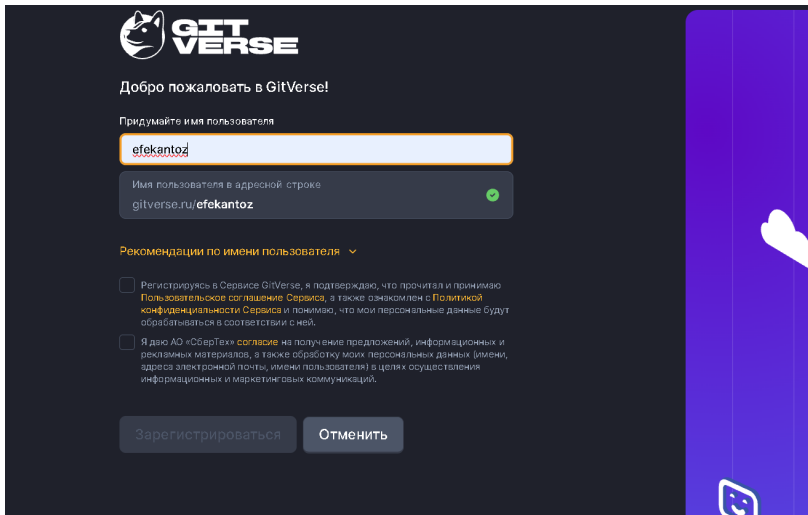
```
Chocolatey installed 1/1 packages.  
See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).  
PS C:\WINDOWS\system32> git config --global user.name "Mehmet efe kantoz"  
PS C:\WINDOWS\system32> git config --global user.email "1032228428@pfur.ru"  
PS C:\WINDOWS\system32> git config --global init.defaultBranch master  
PS C:\WINDOWS\system32> git config --global core.autocrlf input  
PS C:\WINDOWS\system32> git config --global core.safecrlf warn  
PS C:\WINDOWS\system32>
```

Рисунок 4: Настроим utf-8 в выводе сообщений git:

```
Chocolatey installed 1/1 packages.  
See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).  
PS C:\WINDOWS\system32> git config --global user.name "Mehmet efe kantoz"  
PS C:\WINDOWS\system32> git config --global user.email "1032228428@pfur.ru"  
PS C:\WINDOWS\system32> git config --global init.defaultBranch master  
PS C:\WINDOWS\system32> git config --global core.autocrlf input  
PS C:\WINDOWS\system32> git config --global core.safecrlf warn  
PS C:\WINDOWS\system32>
```

Рисунок 5: Зададим имя начальной ветки (будем называть её master):

Настройка gitverse 1



 **GIT
VERSE**

Добро пожаловать в GitVerse!

Придумайте имя пользователя

Имя пользователя в адресной строке
gitverse.ru/efekantoz

Рекомендации по имени пользователя ▾

☐ Регистрируясь в Сервисе GitVerse, я подтверждаю, что прочитал и принимаю **Пользовательское соглашение Сервиса**, а также ознакомлен с **Политикой конфиденциальности Сервиса** и понимаю, что мои персональные данные будут обрабатываться в соответствии с ней.

☐ Я даю АО «СберТех» **согласие** на получение предложений, информационных и рекламных материалов, а также обработку моих персональных данных (имени, адреса электронной почты, имени пользователя) в целях осуществления информационных и маркетинговых коммуникаций.

Рисунок 6: Настройка gitverse:

Настройка gitverse 2

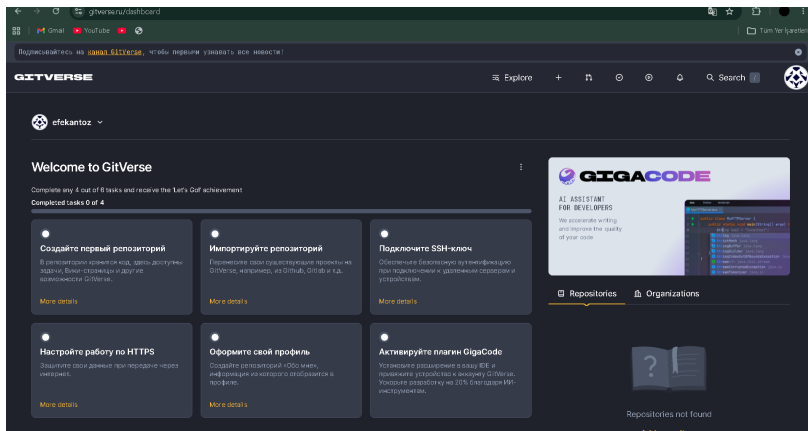
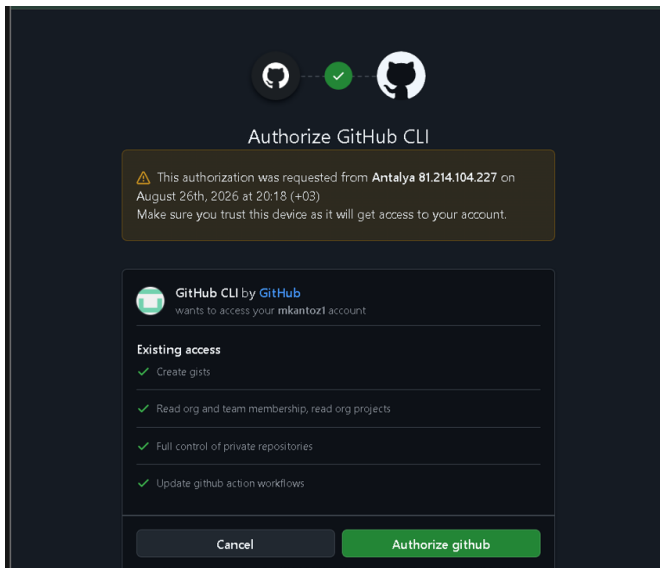


Рисунок 7: Создайте учётную запись на gitverse



Доступ к github

```
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Saving key "~/.ssh/id_ed25519-git" failed: No such file or directory
PS C:\WINDOWS\system32> New-Item -ItemType Directory -Path "$HOME\.ssh" -Force

Directory: C:\Users\uzarb

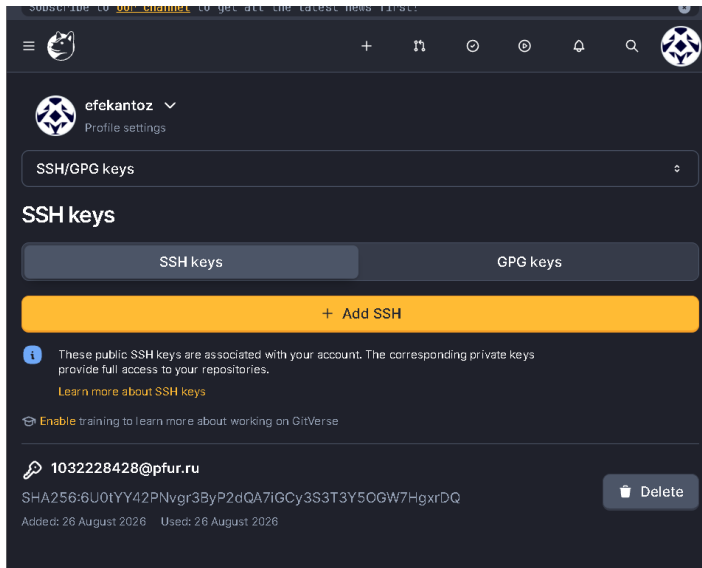
Mode                LastWriteTime         Length Name
----                -
d-----          26.08.2026      20:28         .ssh

PS C:\WINDOWS\system32> ssh-keygen -t ed25519 -f "$HOME\.ssh\id_ed25519-git" -C "1032228428@pfur.ru"
Generating public/private ed25519 key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\uzarb\.ssh\id_ed25519-git
Your public key has been saved in C:\Users\uzarb\.ssh\id_ed25519-git.pub
The key fingerprint is:
SHA256:6U0tYY42PNvgr3ByP2dQA7iGCy3S3T3Y5OGW7HgxrDQ 1032228428@pfur.ru
The key's randomart image is:
+--[ED25519 256]--+
|
| . +
| . o o @o+
| . + +.E=&o o
| . o +S*+=..
|   +oO+.
|   o *.o.
|   = o. o
|   ..o+
|
+----[SHA256]-----+
PS C:\WINDOWS\system32>
```

Добавление ключа ssh

```
PS C:\WINDOWS\system32> ssh-add "$HOME\.ssh\id_ed25519-git"  
Identity added: C:\Users\uzarb\.ssh\id_ed25519-git (1032228428@pfur.ru)  
PS C:\WINDOWS\system32> gh ssh-key add "$HOME\.ssh\id_ed25519-git.pub" --title "1032228428@pfur.ru"  
HTTP 404: Not Found (https://api.github.com/user/keys?per_page=100)  
This API operation needs the "admin:public_key" scope. To request it, run: gh auth refresh -h github.co  
m -s admin:public_key  
PS C:\WINDOWS\system32> gh ssh-key add "$HOME\.ssh\id_ed25519-git.pub" --title "1032228428@pfur.ru" --ty  
pe signing  
HTTP 404: Not Found (https://api.github.com/user/ssh_signing_keys?per_page=100)  
This API operation needs the "admin:ssh_signing_key" scope. To request it, run: gh auth refresh -h gith  
ub.com -s admin:ssh_signing_key  
PS C:\WINDOWS\system32>
```

Рисунок 10: Добавляем ключ ssh-add



```
m -s admin:ssh_signing_key
PS C:\WINDOWS\system32> gh ssh-key add "%HOME\.ssh\id_ed25519-git.pub" --title "1032228428@pfur.ru" --ty
pe signing
HTTP 404: Not Found (https://api.github.com/user/ssh_signing_keys?per_page=100)
This API operation needs the "admin:ssh_signing_key" scope. To request it, run: gh auth refresh -h gith
ub.com -s admin:ssh_signing_key
PS C:\WINDOWS\system32>
```

Рисунок 12: Добавьте ключ в учётную запись github:

ssh-key add

```
! First copy your one-time code: 7611-B7FF
Press Enter to open https://github.com/login/device in your browser...
✓ Authentication complete.
PS C:\WINDOWS\system32> gh ssh-key add "$HOME\.ssh\id_ed25519-git.pub" --title "1032228428@pfur.ru"
✓ Public key added to your account
PS C:\WINDOWS\system32> gh ssh-key add "$HOME\.ssh\id_ed25519-git.pub" --title "1032228428@pfur.ru" --ty
pe signing
✓ Public key added to your account
PS C:\WINDOWS\system32>
```

Рисунок 13: Добавьте ssh-key add:

Генерация GPG ключа

```
Administrator: Windows PowerShell
E-posta adresiniz: 1032228428@pfur.ru
Ön bilgi:
Seçtiğiniz KULLANICI KİMLİĞİ:
"Mehmet efe kantoz <1032228428@pfur.ru>"

(A)d, Ö(n) bilgi, (E)-posta değiştir veya (T)anam/Cı(k)? 0
(A)d, Ö(n) bilgi, (E)-posta değiştir veya (T)anam/Cı(k)? 0
(A)d, Ö(n) bilgi, (E)-posta değiştir veya (T)anam/Cı(k)? k
gpg: Anahtar üretimi iptal edildi.
PS C:\WINDOWS\system32> gpg --full-generate-key
gpg (GnuPG) 2.5.21; Copyright (C) 2026 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Lütfen istediğiniz anahtar türünü seçin:
(1) RSA ve RSA
(2) DSA ve ElGamal
(3) DSA (yalnızca imzalama)
(4) RSA (yalnızca imzalama)
(9) ECC (imza ve şifreleme) *Öntanımlı*
(10) DSA (yalnızca imzalama)
(14) Karttaki var olan anahtar
(16) ECC ve Kyber
Seçiminiz? 1
RSA anahtarları 1024 bit ile 4096 bit arasında olmalıdır.
İstediğiniz anahtar boyutu nedir? (3072) 4096
İstenen anahtar boyutu: 4096 bit
Lütfen anahtarın ne kadar süreyle geçerli olacağını belirtin.
0 = anahtar süresiz geçerli
<n> = anahtar n gün geçerli
<n>w = anahtar n hafta geçerli
<n>m = anahtar n ay geçerli
<n>y = anahtar n yıl geçerli
Anahtar ne kadar süre için geçerli olsun? (0) 0
Anahtarın geçerliliği hiçbir zaman bitmeyecek
Bu doğru mu? (e/h) y

GnuPG'nin anahtarınızı tanımlaması için bir kimlik yapması gerekiyor.

Gerçek adınız: mehmet efe kantoz
E-posta adresiniz: 1032228428@pfur.ru
Ön bilgi:
Seçtiğiniz KULLANICI KİMLİĞİ:
"Mehmet efe kantoz <1032228428@pfur.ru>"
```

```
PS C:\WINDOWS\system32> gpg --list-secret-keys --keyid-format LONG
gpg: güvence veritabanı denetleniyor
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: derinlik: 0 geçerli: 1 imzalı: 0 güvenilir: 0-, 0q, 0n, 0m, 0f, 1u
[keyboxd]
-----
sec   rsa4096/0228F448E965B559 2026-08-26 [SC]
      38F9DF25B166CBE9BF7DF6B90228F448E965B559
uid   [      son derece      ] mehmet efe kantoz <1032228428@pfur.ru>
ssb   rsa4096/D537EE2A7286B900 2026-08-26 [E]
      DF637CDEB0C3A80C2583DFC6D537EE2A7286B900
```

Рисунок 15: Выводим список ключей gpg –list-secret-keys:

Копирование PGP ключа

```
uid [son derece] mehmet efe kantoz <1032228428@ptur.ru>  
ssb rsa4096/D537EE2A7286B900 2026-08-26 [E]  
DF637CDEB0C3A80C2583DFC6D537EE2A7286B900  
  
PS C:\WINDOWS\system32> gpg --armor --export <PGP Fingerprint> | wl-copy
```

Рисунок 16: Скопируйте ваш сгенерированный PGP ключ в буфер обмена:

Добавление GPG в Gitverse

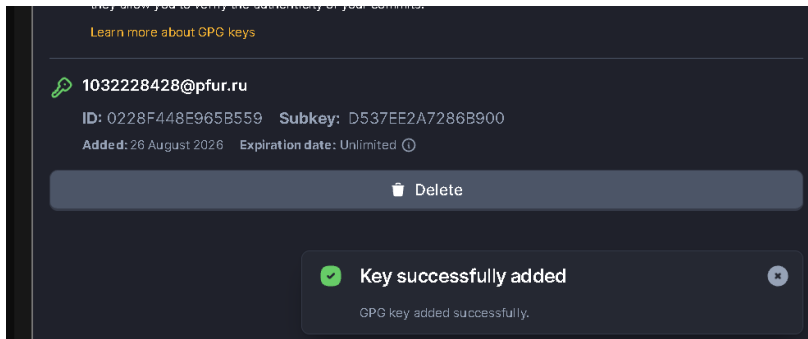


Рисунок 17: Перейдите в настройки Gitverse, нажмите на кнопку Добавить GPG :

Добавление GPG в GitHub

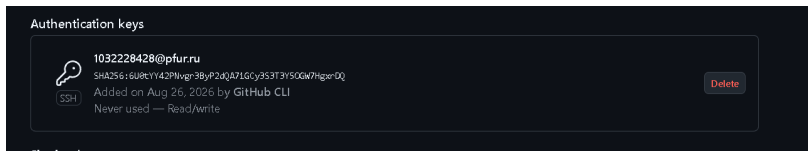


Рисунок 18: Перейдите в настройки GitHub, нажмите на кнопку Добавить GPG :

```
+ FullyQualifiedErrorId : RedirectionNotSupported  
PS C:\WINDOWS\system32> gpg --armor --export 38F9DF25B166CBE9BF7DF6B90228F448E965B559 | Set-Clipboard  
PS C:\WINDOWS\system32> gpg --armor --export 38F9DF25B166CBE9BF7DF6B90228F448E965B559 | Set-Clipboard  
PS C:\WINDOWS\system32> gpg --armor --export 38F9DF25B166CBE9BF7DF6B90228F448E965B559 | Set-Clipboard  
PS C:\WINDOWS\system32> git config --global user.signingkey 38F9DF25B166CBE9BF7DF6B90228F448E965B559  
PS C:\WINDOWS\system32> git config --global commit.gpgsign true  
PS C:\WINDOWS\system32> git config --global user.signingkey <PGP Fingerprint>
```

Рисунок 19: Настройка автоматических подписей коммитов git:

```
gpg: program G:\Program Files\GnuPG\bin\gpg.exe
PS C:\WINDOWS\system32> choco install quarto
Chocolatey v2.7.4
Installing the following packages:
quarto
By installing, you accept licenses for the packages.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading quarto 1.10.18... 100%

quarto v1.10.18 [Approved]
quarto package files install completed. Performing other installation steps.
The package quarto wants to run 'chocolateyinstall.ps1'.
Note: If you don't run this script, the installation will fail.
Note: To confirm automatically next time, use '-y' or consider:
choco feature enable -n allowGlobalConfirmation
Do you want to run the script?([Y]es/[A]ll scripts/[N]o/[P]rint):
```

Рисунок 20: choco install quarto:

Установка Node, yarn, pnpm

```
! You were already logged in to this account
PS C:\WINDOWS\system32> pnpm add -g commitizen
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +148
+++++
Packages are hard linked from the content-addressable store to the virtual store.
  Content-addressable store is at: C:\Users\uzarb\AppData\Local\pnpm\store\v11
  Virtual store is at:           ../../Users/uzarb/AppData/Local/pnpm/store/v11/links
Progress: resolved 167, reused 0, downloaded 148, added 148, done
Downloading @typescript/typescript-win32-x64@7.0.2: 9,77 MB/9,77 MB, done

global:
+ commitizen 4.3.2

Done in 20.5s using pnpm v11.24.0
PS C:\WINDOWS\system32>

PS C:\WINDOWS\system32> yarn global add commitizen
yarn global v1.22.22
(node:22176) [DEP0169] DeprecationWarning: `url.parse()` behavior is not standardized and prone to error
s that have security implications. Use the WHATWG URL API instead. CVEs are not issued for `url.parse()`
vulnerabilities.
(Use `node --trace-deprecation ...` to show where the warning was created)
[1/4] Resolving packages...
warning commitizen > glob@7.2.3: Old versions of glob are not supported, and contain widely publicized s
ecurity vulnerabilities, which have been fixed in the current version. Please update. Support for old ve
rsions may be purchased (at exorbitant rates) by contacting i@izs.me
```

```
Done in 17.79s.  
PS C:\WINDOWS\system32> pnpm add -g cz-customizable  
WARN 3 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6, rimraf@2.6.3  
Packages: +55  
+++++  
Packages are hard linked from the content-addressable store to the virtual store.  
Content-addressable store is at: C:\Users\uzarb\AppData\Local\pnpm\store\v11  
Virtual store is at:           ../../Users/uzarb/AppData/Local/pnpm/store/v11/links  
Progress: resolved 55, reused 25, downloaded 30, added 30, done  
  
global:  
+ cz-customizable 7.5.4  
  
Done in 6.8s using pnpm v11.24.0  
PS C:\WINDOWS\system32>
```

Рисунок 21: pnpm add -g cz-customizable:

standard-version

```
Administrator: Windows PowerShell
warning standard-version > conventional-changelog > conventional-changelog-eslint@3.0.9: This preset is deprecated. Please use conventional-changelog-conventionalcommits or conventional-changelog-angular instead.
warning standard-version > conventional-changelog > conventional-changelog-eslint > @1.5.1: You or someone you depend on is using Q, the JavaScript Promise library that gave JavaScript developers strong feelings about promises. They can almost certainly migrate to the native JavaScript promise now. Thank you! I literally everyone for joining me in this bet against the odds. Be excellent to each other.

(For a CapTP with native promises, see @endo/eventual-send and @endo/captp)
warning standard-version > conventional-changelog > conventional-changelog-jshint@2.0.9: This preset is deprecated. Please use conventional-changelog-conventionalcommits or conventional-changelog-angular instead.
warning standard-version > conventional-changelog > conventional-changelog-jshint > @1.5.1: You or someone you depend on is using Q, the JavaScript Promise library that gave JavaScript developers strong feelings about promises. They can almost certainly migrate to the native JavaScript promise now. Thank you! I literally everyone for joining me in this bet against the odds. Be excellent to each other.

(For a CapTP with native promises, see @endo/eventual-send and @endo/captp)
[2/4] Fetching packages...
[3/4] Linking dependencies...
warning "commitizen > cz-conventional-changelog > @commitlint/load > cosmiconfig-typescript-loader@6.3.0" has unmet peer dependency "@types/node@"*.
warning "commitizen > cz-conventional-changelog > @commitlint/load > cosmiconfig-typescript-loader@6.3.0" has unmet peer dependency "typescript@>5".
warning Workspaces can only be enabled in private projects.
warning Workspaces can only be enabled in private projects.
[4/4] Building fresh packages...
success Installed "standard-version@9.5.0" with binaries:
  - standard-version
Done in 7.92s.
PS C:\WINDOWS\system32>
```

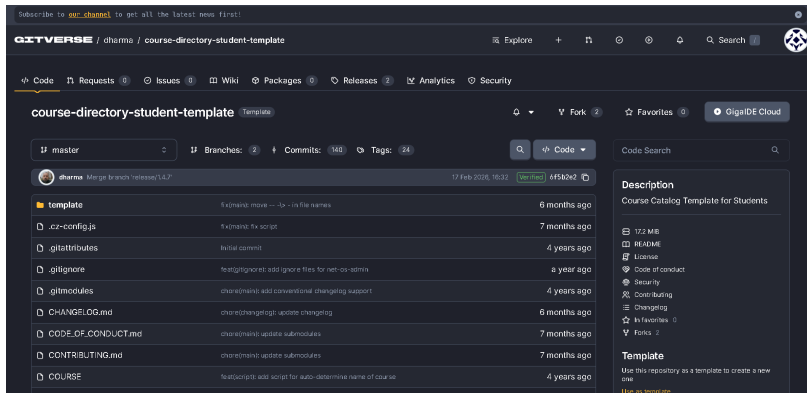


Рисунок 23: Репозиторий GitVerse:

The screenshot shows the GitHub interface for a repository named '2026-1-study-simulation-modeling' owned by 'mkantoz1'. The repository is public and has 1 branch (master) and 1 tag. The commit history shows 2 commits. The file list includes folders 'labs' and 'template', and files '.cz-config.js', '.gitattributes', '.gitignore', '.gitmodules', 'CODE_OF_CONDUCT.md', 'CONTRIBUTING.md', 'COURSE', and 'LICENSE'. The commit messages for the files are: 'feat(main): make course structure' for 'labs', 'template', and 'COURSE'; and 'Initial commit' for the others.

mkantoz1 / 2026-1-study-simulation-modeling

Public

master 1 Branch 1 Tag

Go to file Add file Code

mkantoz1 feat(main): make course structure 6969cf6 · yesterday 2 Commits

labs	feat(main): make course structure	yesterday
template	Initial commit	yesterday
.cz-config.js	Initial commit	yesterday
.gitattributes	Initial commit	yesterday
.gitignore	Initial commit	yesterday
.gitmodules	feat(main): make course structure	yesterday
CODE_OF_CONDUCT.md	Initial commit	yesterday
CONTRIBUTING.md	Initial commit	yesterday
COURSE	feat(main): make course structure	yesterday
LICENSE	Initial commit	yesterday

```
Receiving objects: 100% (41/41), 25.31 KiB | 1.81 MiB/s, done.  
Resolving deltas: 100% (1/1), done.  
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling> make help  
'Usage:'  
Sistem belirtilen dosyayı bulamıyor.  
make: *** [Makefile:10: help] Error 1  
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling>
```

Рисунок 25: Посмотреть доступные цели make help:

```
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling> git clone https://git-verse.ru/efekantoz/2026-1-study-simulation-modeling.git .
Cloning into '.'...
remote: Enumerating objects: 41, done.
remote: Counting objects: 100% (41/41), done.
remote: Compressing objects: 100% (40/40), done.
remote: Total 41 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (41/41), 25.31 KiB | 1.81 MiB/s, done.
Resolving deltas: 100% (1/1), done.
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling>
```

Рисунок 26: Укажите название для вашего нового проекта:

```
Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master)
$ make list
practical-scientific-writing Computer Skills for Scientific Writing
net-admin Администрирование локальных сетей
net-os-admin Администрирование сетевых подсистем
arch-pc Архитектура ЭВМ
sciprogram-intro Введение в научное программирование
netcybersec Защита сетей и кибербезопасность
simulation-modeling Имитационное моделирование
infosec Информационная безопасность
computer-practice Компьютерный практикум по статистическому анализу данных
mathsec Математические основы защиты информации и информационной безопасности
mathmod Математическое моделирование
security-modeling Методы математического моделирования в кибербезопасности
simulation-networks Моделирование сетей передачи данных
sciprogram Научное программирование
os-intro Операционные системы
os2 Основы администрирования операционных систем
infosec-intro Основы информационной безопасности
nettech Сетевые технологии
```

Рисунок 27: Посмотреть список доступных курсов (make list):

```
Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master)
$ make prepare
Submodule 'template/presentation' (https://gitverse.ru/dharma/academic-presentation-markdown-template.git) registered for path 'template/presentation'
Submodule 'template/report' (https://gitverse.ru/dharma/academic-laboratory-report-template.git) registered for path 'template/report'
Cloning into 'C:/Users/uzarb/work/study/2026-1/2026-1==study-simulation-modeling/template/presentation'...
Cloning into 'C:/Users/uzarb/work/study/2026-1/2026-1==study-simulation-modeling/template/report'...
Submodule path 'template/presentation': checked out '60a6091ebdbb01a8d98f2c5fae0f98dc63ad031'
Submodule path 'template/report': checked out '708b8c51acd94d4b78e7634198f221e5dec6b355'
Synchronizing submodule url for 'template/presentation'
Synchronizing submodule url for 'template/report'

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master)
$
```

Рисунок 28: Инициализируйте курс: echo simulation-modeling >COURSE make prepare:

```
Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master)  
$ git add .
```

Рисунок 29: Отправьте файлы на сервер (git add):

Git push

```

MINGW64~/c/Users/uzarb/work/study/2026-1/2026-1==study-simulation-modeling
Targets:
list          List of courses
prepare       Generate directories structure
submodule     Update submodules

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master)
$ make list
practical-scientific-writing Computer Skills for Scientific Writing
net-admin                   Администрирование локальных сетей
net-os-admin                Администрирование сетевых подсистем
arch-pc                     Архитектура ЗЕМ
sciprogram-intro            Введение в научное программирование
netcybersec                 Защита сетей и кибербезопасность
simulation-modeling          Имитационное моделирование
infosec                     Информационная безопасность
computer-practice           Компьютерный практикум по статистическому анализу данных
mathsec                     Математические основы защиты информации и информационной безоп
асности
mathmod                     Математическое моделирование
security-modeling            Методы математического моделирования в кибербезо
simulation-networks          Моделирование сетей передачи данных
sciprogram                  Научное программирование
os-intro                    Операционные системы
os2                          Основы администрирования операционных систем
infosec-intro               Основы информационной безопасности
nettech                     Сетевые технологии

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master)
$ echo simulation-modeling > COURSE

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master)
$ make prepare
Submodule 'template/presentation' (https://gitverse.ru/dharma/academic-presentation-template.git) registered for path 'template/presentation'
Submodule 'template/report' (https://gitverse.ru/dharma/academic-laboratory-report-template.git) registered for path 'template/report'
Cloning into 'C:/Users/uzarb/work/study/2026-1/2026-1==study-simulation-modeling/template/presentation'...
Cloning into 'C:/Users/uzarb/work/study/2026-1/2026-1==study-simulation-modeling/template/report'...
Submodule path 'template/presentation': checked out '60a6091ebdbb01a8d98f2c5fae0f98dc63ad031'
Submodule path 'template/report': checked out '708b8c51acd94d4b78e7634198f221e5dec6335'
Synchronizing submodule url for 'template/presentation'
Synchronizing submodule url for 'template/report'

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master)
$ git add.
git: 'add.' is not a git command. See 'git --help'.

The most similar command is

```



OpenPGP gizli anahtarının kilidini açmak için anahtar parolasını girin:
"mehmet efe kantoz <1032228428@pfur.ru>"
4096 bit RSA anahtarı, 0228F448E9658559 kimliği,
2026-08-26 tarihinde oluşturuldu.

Anahtar Parolası:

Anahtar parolası  (3/3 dene)

OKCancel

Push на github

```
Efe Kantoz@PC MINGW64 ~  
$ cd ~/work/study/2026-1/2026-1==study-simulation-modeling  
  
Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master)  
$ git remote set-url origin https://67e42b5eaaebe1ee271e29974fe77d7ab8c79b42@gitverse.ru/efekantoz/2026-1-study-simulation-modeling.git  
  
Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master)  
$ |
```

Рисунок 31: Запуште репозиторий на github:

Конфигурация git-flow

```
Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master)
$ git flow init

Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [C:/Users/uzarb/work/study/2026-1/2026-1==study-simulation-modeling/.git/hooks]

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (develop)
N$
```

Рисунок 32: Конфигурация git-flow:

```
Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master)
$ git flow init

Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [C:/Users/uzarb/work/study/2026-1/2026-1==study-simulation-modeling/.git/hooks]

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (develop)
N$
```

Рисунок 33: Проверьте, что Вы на ветке develop:

Загрузка репозитория

```
Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (develop)
$ git branch
* develop
  master

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (develop)
$ git push -u --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: You can create a PR using the link below:
remote:   https://gitverse.ru/efekantoz/2026-1-study-simulation-modeling/compare?baseBranch=master&headBranch=develop
remote:
remote: . Processing 1 references
remote: Processed 1 references in total
To https://gitverse.ru/efekantoz/2026-1-study-simulation-modeling.git
 * [new branch]      develop -> develop
branch 'master' set up to track 'origin/master'.
branch 'develop' set up to track 'origin/develop'.

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (develop)
$
```

Рисунок 34: Загрузите весь репозиторий в хранилище:

```
branch 'develop' set up to track 'origin/develop' .  
Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (develop)  
$ git flow release start 1.0.0  
Switched to a new branch 'release/1.0.0'  
  
Summary of actions:  
- A new branch 'release/1.0.0' was created, based on 'develop'  
- You are now on branch 'release/1.0.0'  
  
Follow-up actions:  
- Bump the version number now!  
- Start committing last-minute fixes in preparing your release  
- When done, run:  
  
    git flow release finish '1.0.0'
```

Рисунок 35: Создадим релиз с версией 1.0.0:

```
Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (release/1.0.0)
$ standard-changelog --first-release
bash: standard-changelog: command not found

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (release/1.0.0)
$ standard-changelog --first-release
bash: standard-changelog: command not found

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (release/1.0.0)
$
```

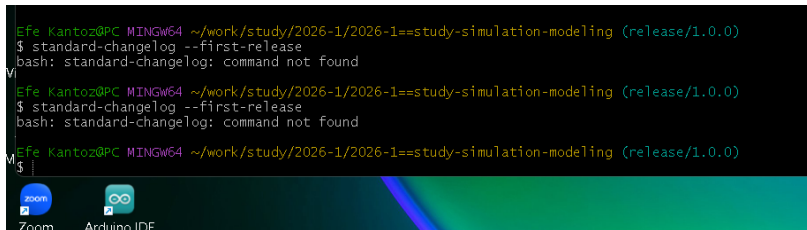


Рисунок 36: Создадим журнал изменений

```
Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (release/1.0.0)
$ git add CHANGELOG.md
Mgit commit -am 'chore(site): add changelog'
[release/1.0.0 c703f0b] chore(site): add changelog
1 file changed, 5 insertions(+)
create mode 100644 CHANGELOG.md

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (release/1.0.0)
```

Рисунок 37: Добавим журнал изменений в индекс

```
MINGW64~/work/study/2026-1/2026-1==study-simulation-modeling
remote: . Processing 1 references
remote: Processed 1 references in total
To https://gitverse.ru/efekantoz/2026-1-study-simulation-modeling.git
 * [new branch]      develop -> develop
branch 'master' set up to track 'origin/master'.
branch 'develop' set up to track 'origin/develop'.

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (develop)
$ git flow release start 1.0.0
Switched to a new branch 'release/1.0.0'

Summary of actions:
- A new branch 'release/1.0.0' was created, based on 'develop'
- You are now on branch 'release/1.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.0.0'

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (release/1.0.0)
$ standard-changelog --first-release
bash: standard-changelog: command not found

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (release/1.0.0)
$ standard-changelog --first-release
bash: standard-changelog: command not found

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (release/1.0.0)
$ npm install -g standard-changelog
standard-changelog --first-release

added 17 packages in 12s

Switched to branch 'master'
Your branch is up to date with 'origin/master'.
hint: Waiting for your editor to close the file...
```

```
branch 'develop' set up to track 'origin/develop'.
branch 'master' set up to track 'origin/master'.
branch 'release/1.0.0' set up to track 'origin/release/1.0.0'.

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master|MERGING)
$ git push --all
git push --tags
Everything up-to-date
Everything up-to-date

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master|MERGING)
$ mkdir -p ../release
cp CHANGELOG.md ../release
gh release create v1.0.0 -F ../release/CHANGELOG.md
https://github.com/mkantoz1/2026-1-study-simulation-modeling/releases/tag/v1.0.0

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master|MERGING)
$
```

Рисунок 39: Отправим данные на github

Копирование CHANGELOG

```
Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master|MERGING)
$ mkdir -p ../release
$ cp CHANGELOG.md ../release
$ gh release create v1.0.0 -F ../release/CHANGELOG.md
https://github.com/mkantozi/2026-1-study-simulation-modeling/releases/tag/v1.0.0
Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master|MERGING)
$ git commit --no-edit
[master 77b67bd] Merge branch 'release/1.0.0' # Please enter a commit message to explain why this me
rge is necessary, # especially if it merges an updated upstream into a topic branch. # # Lines start
ing with '#' will be ignored, and an empty message aborts # the commit.
Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master)
$
```

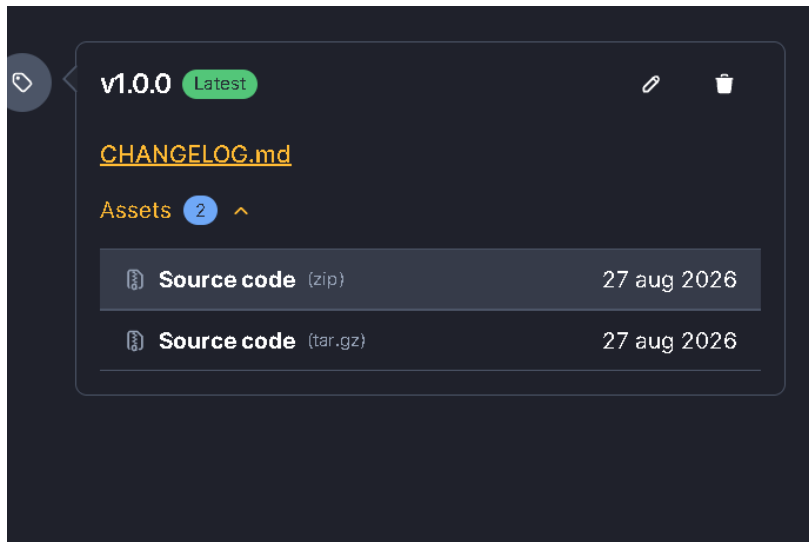
Рисунок 40: Скопируем CHANGELOG.md в каталог release:

Push all

```
Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master)
$ git push --all
git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 1.05 KiB | 1.05 MiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: . Processing 1 references
remote: Processed 1 references in total
To https://gitverse.ru/efekantoz/2026-1-study-simulation-modeling.git
   6969cf6..77b67bd master -> master
Everything up-to-date

Efe Kantoz@PC MINGW64 ~/work/study/2026-1/2026-1==study-simulation-modeling (master)
$
```

Рисунок 41: Push all:



```
No package found matching input criteria.  
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01> _
```

Рисунок 43: Создадим LAB01

```
Vis + FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01> Get-ChildItem -Path "$env:LOCALAPPDATA" -
Filter "julia.exe" -Recurse -ErrorAction SilentlyContinue | Select-Object -ExpandProperty FullName
C:\Users\uzarb\AppData\Local\Programs\Julia-1.12.7\bin\julia.exe
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01> C:\Users\uzarb\AppData\Local\Programs\Jul
ia-1.12.7\bin\julia.exe

Documentation: https://docs.julialang.org
Type "?" for help, "]" for Pkg help.
Version 1.12.7 (2026-08-15)
Official https://julialang.org release

julia>
```

Рисунок 44: Откройте терминал и запустите Julia:

```
julia> Pkg.add("DrWatson")
   Project No packages added to or removed from `C:\Users\uzarb\work\study\2026-1\2026-1==study-simula
tion-modeling\labs\lab01\project\Project.toml`
   Manifest No packages added to or removed from `C:\Users\uzarb\work\study\2026-1\2026-1==study-simula
tion-modeling\labs\lab01\project\Manifest.toml`
julia>
```

Рисунок 45: В REPL выполните:

Основные пакеты

```
Updating C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project\proj
ect\Project.toml
[0c46a032] + DifferentialEquations v8.1.0
Updating C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project\proj
ect\Manifest.toml
[47edcb42] + ADTypes v1.24.0
[14f7f29c] + AMD v0.5.3
[7d9f7c33] + Accessors v0.1.45
[79e6a3ab] + Adapt v4.7.0
[4fba245c] + ArrayInterface v7.30.0
[b2a6c25c] + BinaryHeaps v1.1.0
[70df07ce] + BracketingNonlinearSolve v1.12.6
[38540f10] + CommonSolve v0.2.14
[bbf7d656] + CommonSubexpressions v0.3.1
[34da2185] + Compat v4.18.1
[a33af91c] + CompositionsBase v0.1.2
[2569d6c7] + ConcreteStructs v0.2.8
[187b0558] + ConstructionBase v1.6.0
[a8cc5b0e] + Crayons v4.2.0
[9a962f9c] + DataAPI v1.16.0
[e2d170a0] + DataValueInterfaces v1.0.0
[2b5f629d] + DiffEqBase v7.18.2
[163ba53b] + DiffResults v1.1.0
[b552c78f] + DiffRules v1.16.0
[0c46a032] + DifferentialEquations v8.1.0
[a0c0ee7d] + DifferentiationInterface v0.7.21
[ffbed154] + DocStringExtensions v0.9.5
[4e289a0a] + EnumX v1.0.7
[f151be2c] + EnzymeCore v0.8.21
[e2ba6199] + ExprTools v0.1.11
[7034ab61] + FastBroadcast v1.4.0
[9aa1b823] + FastClosures v0.3.2
[a4df4552] + FastPower v1.5.0
[64ca27bc] + FindFirstFunctions v3.2.1
[6a86dc24] + FiniteDiff v2.33.0
[f6369f11] + ForwardDiff v1.4.5
[069b7b12] + FunctionWrappers v1.1.3
[77dc65aa] + FunctionWrappersWrappers v1.13.0
```

```
julia> include("scripts/test_setup.jl")
✓ Проект активирован: C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project

Проверка пакетов:
  ✓ DrWatson
  ✓ DifferentialEquations
  ✓ Plots
  ✓ DataFrames
  ✓ CSV
  ✓ JLD2
  ✓ Literate
  ✓ IJulia
  ✓ BenchmarkTools
  ✓ Quarto

Структура проекта:
Корень: C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project
Данные: C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project\data
Скрипты: C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project\src
Графика: C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project\plots

julia>
```

Рисунок 47: Запустим:

```
using DrWatson
julia> open("scripts/01_exponential_growth.jl", "w") do f
    write(f, """
    # # Экспоненциальный рост
    # **Цель:** Исследовать решение уравнения  $du/dt = \alpha u$ .

    # ## Инициализация проекта и загрузка пакетов
    using DrWatson
    @quickactivate "project"

    using DifferentialEquations
    using Plots
    using DataFrames
    using JLD2

    script_name = "01_exponential_growth"
    mkpath(plotsdir(script_name))
    mkpath(datadir(script_name))

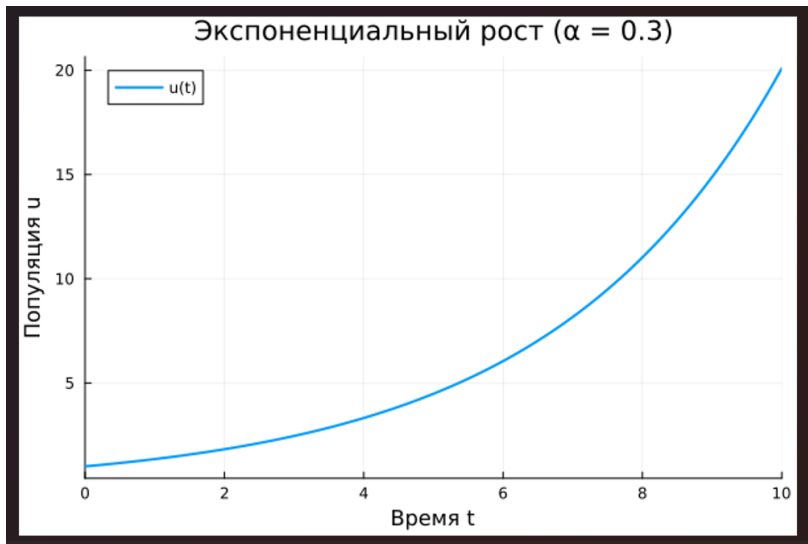
    # ## Определение модели
    # Уравнение экспоненциального роста:
    #  $du/dt = \alpha u$ ,  $u(0) = u_0$ 

    function exponential_growth!(du, u, p, t)
         $\alpha = p$ 
         $du[1] = \alpha * u[1]$ 
    end

    # ## Первый запуск с параметрами по умолчанию
    # Зададим начальные параметры:
    u0 = [1.0]          # начальная популяция
     $\alpha = 0.3$         # скорость роста
    tspan = (0.0, 10.0) # временной интервал

    prob = ODEProblem(exponential_growth!, u0, tspan,  $\alpha$ )
    sol = solve(prob, Tsit5(), saveat=0.1)

    # ## Визуализация результатов
    # Построим график решения:
    plot(sol, label="u(t)", xlabel="Время t", ylabel="Популяция u",
        title="Экспоненциальный рост ( $\alpha = \backslash\alpha$ ", lw=2, legend=:topleft)
```



```
julia> using Literate

julia> Literate.markdown("scripts/01_exponential_growth.jl", "docs", execute=true)
[ Info: generating markdown page from `C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modelin
\labs\lab01\project\scripts\01_exponential_growth.jl`
[ Info: writing result to `C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01
\project\docs\01_exponential_growth.md`
"C:\\Users\\uzarb\\work\\study\\2026-1\\2026-1==study-simulation-modeling\\labs\\lab01\\project\\docs\\0
1_exponential_growth.md"

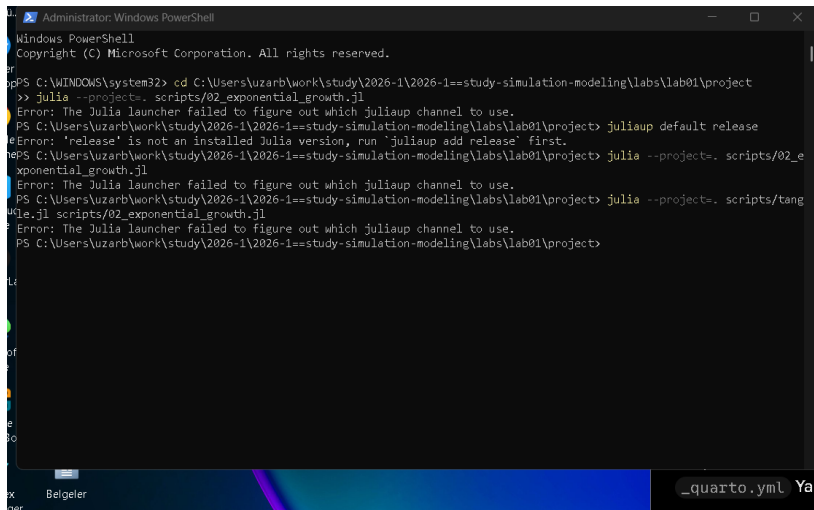
julia> quarto render report
ERROR: ParseError:
# Error @ REPL[22]:1:7
quarto render report
#      └─ extra tokens after end of expression

Stacktrace:
 [1] top-level scope
      @ REPL:1

julia>
```

Рисунок 50: Добавьте в файл описание в духе литературного программирования:

Выполнение программы



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> cd C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project
>> julia --project=. scripts/02_exponential_growth.jl
Error: The Julia launcher failed to figure out which juliaup channel to use.
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project> juliaup default release
Error: 'release' is not an installed Julia version, run 'juliaup add release' first.
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project> julia --project=. scripts/02_e
xponential_growth.jl
Error: The Julia launcher failed to figure out which juliaup channel to use.
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project> julia --project=. scripts/tang
le.jl scripts/02_exponential_growth.jl
Error: The Julia launcher failed to figure out which juliaup channel to use.
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project>
```

Рисунок 51: Выполните программу `julia --project=. scripts/02_exponential_growth.jl`:

Поддержка кода Julia

```
Administrator: Windows PowerShell

From https://gitverse.ru/efekantoz/2026-1-study-simulation-modeling
* branch      master      -> FETCH_HEAD
Already up to date.
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project> git merge develop
error: Your local changes to the following files would be overwritten by merge:
  labs/lab01/project/.gitattributes labs/lab01/project/.github/workflows/CI.yml labs/lab01/project/.gitignore labs/lab01/project/Project.toml labs/lab01/project/README.md labs/lab01/project/docs/01_exponential_growth.md labs/lab01/project/project/.gitattributes labs/lab01/project/project/.github/workflows/CI.yml labs/lab01/project/project/.gitignore labs/lab01/project/project/Project.toml labs/lab01/project/project/README.md labs/lab01/project/project/scripts/intro.jl labs/lab01/project/project/scripts/test_setup.jl labs/lab01/project/project/src/dummy_src_file.jl labs/lab01/project/project/test/runtests.jl labs/lab01/project/scripts/01_exponential_growth.jl labs/lab01/project/scripts/intro.jl labs/lab01/project/scripts/test_setup.jl labs/lab01/project/src/dummy_src_file.jl labs/lab01/project/test/runtests.jl
<stdin>:60: trailing whitespace.
*.app binary
<stdin>:116: trailing whitespace.
*.eps binary
<stdin>:159: trailing whitespace.
*.war binary
<stdin>:187: trailing whitespace.
*.mn binary
<stdin>:212: trailing whitespace.
# (among others).
warning: squelched 10 whitespace errors
warning: 15 lines add whitespace errors.
Merge with strategy ort failed.
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project> git push origin master
Everything up-to-date
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project> git checkout develop
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project> git add .
>> git commit -m "Lab 01 hazirlik"
>> git push origin develop
On branch develop
Your branch is up to date with 'origin/develop'.

nothing to commit, working tree clean
Everything up-to-date
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project> git checkout master
>> git stash
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
No local changes to save
PS C:\Users\uzarb\work\study\2026-1\2026-1==study-simulation-modeling\labs\lab01\project> git merge develop
>> git push origin master
```

Описываются проведённые действия, в качестве иллюстрации даётся ссылка на иллюстрацию (рис. 1).

Раздел 6

Выводы

Здесь кратко описываются итоги проделанной работы.

Раздел 7

Список литературы

Список литературы